



温州肯恩大学
WENZHOU-KEAN UNIVERSITY

2026SPW GMBA 5885 W02
Strategic Mgt Global Info Sys

Game Theory and Strategic Thinking in Digital Markets

Dawei Chen

Day 5 · 16 May 2026

Where We Are

● Day 1 PM	What IS Is	Definition, core activities, sociotechnical view of IS
● Day 2	How IS Creates Value	IS Strategy Triangle, Five Forces, Value Chain, RBV
● Day 3 AM	Digital Transformation	How IS reshapes value creation, operations, and competition
● Day 3 PM	Who IS Affects	Ethical and social issues: info right, accountability, property right, system quality, quality of life
● Day 4 AM	How AI Fits In	AI in organizations: business value, adoption, human–AI role allocation
● Day 4 PM	How to Work with AI	4D AI Fluency Framework, prompts, skills, agents
● Day 5	How to Think Strategically	<i>Game theory and strategic thinking in digital markets</i>

Today's Roadmap

1

AM-1 (09:00–10:15)

What Is a Game?

- Interdependence as the key
- Four building blocks
- Play four games

2

AM-2 (10:30–11:45)

Finding Equilibrium

- Pre-Nash equilibrium
- Nash equilibrium (NE)
- Applications

3

PM-1 (13:15–14:30)

Sequential Games

- Post-Nash equilibrium (SPNE)
- Bargaining and negotiation
- Credibility and threats

4

PM-2 (14:45–16:00)

Strategic Moves

- Commitment
- Credibility
- Information asymmetry

01

What Is a Game?

- Interdependence as the key
- Four building blocks
- Play four classroom games

What Makes a Decision “Strategic”?

Scenario A

“I am looking for the best cinema in Wenzhou for tonight's movie.”



VS.

Scenario B

“I am looking for the best cinema in Wenzhou, yet avoiding my ex-partner.”



The Difference: Interdependence

- In Scenario A, you optimize against nature (which cinema has better seats, sound, etc.).
- In Scenario B, your best choice depends on what **another person** chooses — and vice versa.
- This **mutual dependence** is the defining feature of a **strategic** situation — the focus of game theory.

Is This a Game?

In a game, outcome depends on both your decision and others' – but you can't control theirs.

Situation	Game?	Why
Price competition between Coca-Cola and Pepsi	?	<i>Each firm's pricing affects the other's profit</i>
Two students deciding whether to study or free-ride in a group project	?	<i>Each student's payoff depends on the other's effort</i>
Negotiating salary with a job candidate	?	<i>Each side responds strategically to the other's demands</i>
Meituan vs Ele.me competing for restaurant merchants in a new city	?	<i>Subsidies, exclusive deals, and pricing all depend on rival's moves</i>
Choosing whether to carry an umbrella based on weather forecast	?	<i>Outcome depends on nature, not another strategic player</i>
Rolling a dice in a casino	?	<i>Chance, not strategic interaction</i>

Game Theory: Origins and Relevance

Game theory was developed in the early 1940s — initially in connection with military strategy during the Cold War era — as the formal study of rational decisions in **strategic situations** where **conflict coexists with mutual dependence**.

Such situations are common:



Within firms

Resource allocation,
team incentives, internal
competition for budgets



Between firms

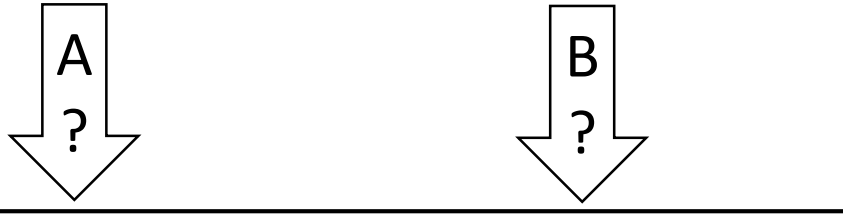
Pricing wars, market entry,
collusion vs. competition,
platform competition



In everyday life

Negotiations, group projects,
social coordination, political
strategy

Motivating Story: The Location Game



- Two street vendors (A and B) on a linear beach road
- Customers are **uniformly** spread along the road and go to the **nearest** vendor.
- **Challenge:** *Where should the vendors locate? What is the logic?*



- ☐ Horizontal: price
- ☐ Vertical: quality

Equilibrium: Both cluster at the center — moving toward the middle always captures more customers.

This is Hotelling's model of **spatial competition**, explaining *why Starbucks and Luckin cluster on the same streets, candidates converge to the center, and competing apps adopt similar features.*

Motivating Story: The Location Game

Analogues in Real Life?

Gas stations, airlines scheduling their flights,
election manifestos between the political parties

- Two 2-yuan bread stores sit side by side on 南柳小吃街.
- Store A has a long queue; Store B has none. Why?
- What would you recommend to Store B? To store A?

2-yuan Bread War on 南柳小吃街



Fundamentals of Games

Every strategic situation can be described by specifying these four elements.



Players

Rational, payoff-maximizing agents:
firms, governments,
individuals



Actions / Strategies

What each player
can choose:
prices, investments,
entry/exit



Preferences / Payoffs

How players rank
outcomes:
ordinal (ranking), not
cardinal (exact numbers)



Information

What each player knows
when choosing:
perfect vs imperfect,
complete vs incomplete

Modeling a Game: Prisoner's Dilemma

The Story

Two suspects are arrested after a joint crime and held in separate cells. The police lack sufficient evidence for a conviction on the main charge.

The prosecutor visits each suspect and makes the same offer:

- *"If you confess and your partner stays silent, you go free and your partner gets 10 years."*
- *"If both of you confess, you each get 5 years."*
- *"If neither confesses, we can only charge you both with a minor offense — 1 year each."*

1 Players

Suspect A and Suspect B

Both are rational and self-interested

Each wants to minimize own prison time

2 Actions / Strategies

Each suspect has two choices:

- Deny (stay silent, cooperate with partner)
- Confess (betray partner, defect)

They choose simultaneously — no communication.

Modeling a Game: Prisoner's Dilemma

		Suspect B	
		Deny	Confess
Suspect A	Deny	1 yr, 1 yr = (2, 2)	10 yr, free = (0, 3)
	Confess	free, 10 yr = (3, 0)	5 yr, 5 yr = (1, 1)

Payoff mapping: free = 3 (best), 1 yr = 2, 5 yr = 1, 10 yr = 0 (worst). *Ordinal ranking — only the order matters.*

3 Information

Each suspect knows:

- ✓ The available strategies (Deny or Confess)
- ✓ The payoff structure (prison sentences for each outcome)

X What the other suspect actually chose — they are in separate cells

4 Preferences / Payoffs

Each suspect ranks outcomes (best to worst):

- 1st: I confess, partner denies → go free (best)
- 2nd: Both deny → 1 year each (good)
- 3rd: Both confess → 5 years each (bad)
- 4th: I deny, partner confesses → 10 years (worst)**

Two Key Distinctions

1. Cooperative vs. Non-cooperative

Non-cooperative: Each player decides independently. Explicit collusion is not allowed.

Cooperative: Coalitions can enforce agreements (e.g., trade negotiations). Collusion is attractive but often illegal.

→ *In the corporate world, we focus on non-cooperative games. But cooperative outcomes can emerge.*

2. Static (Simultaneous) vs. Dynamic (Sequential)

Static: Players choose simultaneously (or without knowing others' choices). Examples: rock-paper-scissors, sealed-bid auctions.

Dynamic: Players move in sequence, observing earlier moves. Examples: chess, market entry decisions.

→ What matters is **information**, not physical order.

Let's Play the Games



Each group: Choose Red or Blue as player A, and choose Red or Blue as player B.

Scoring: For each game, each group gets 4 pairings as A (against the other four groups' B choices) and 4 pairings as B (against the other four groups' A choices) = 8 payoff values per group per game. Sum them up.

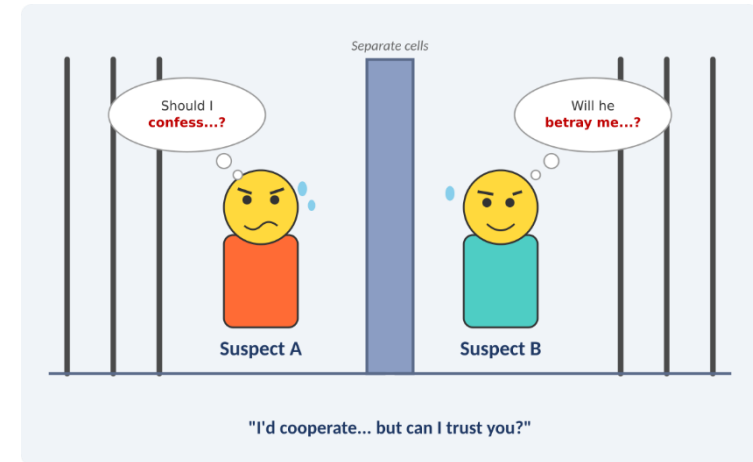
Highest total wins.

Let's Play Games 1

Prisoner's Dilemma

Red → Deny (Cooperate) Blue → Confess (Defect)

		Player B	
		Red	Blue
Player A	Red	(2, 2)	(0, 3)
	Blue	(3, 0)	(1, 1)



Think about the players' preferences

For Player A:

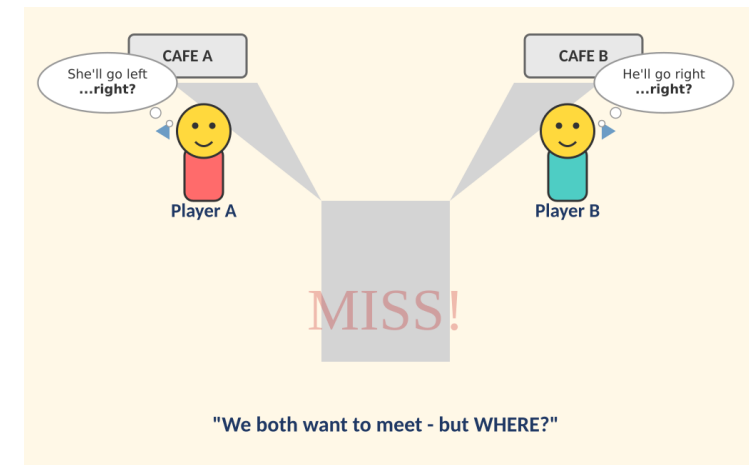
- 1) A deny + B confess → worst for A
- 2) A deny + B deny → good for both
- 3) A confess + B confess → mediocre
- 4) A confess + B deny → best for A

Let's Play Games 2

Coordination Game

Red → Left / Up Blue → Right / Down

		Player B	
		Red	Blue
Player A	Red	(2, 2)	(0, 0)
	Blue	(0, 0)	(1, 1)



Think about the players' preferences

For Player A:

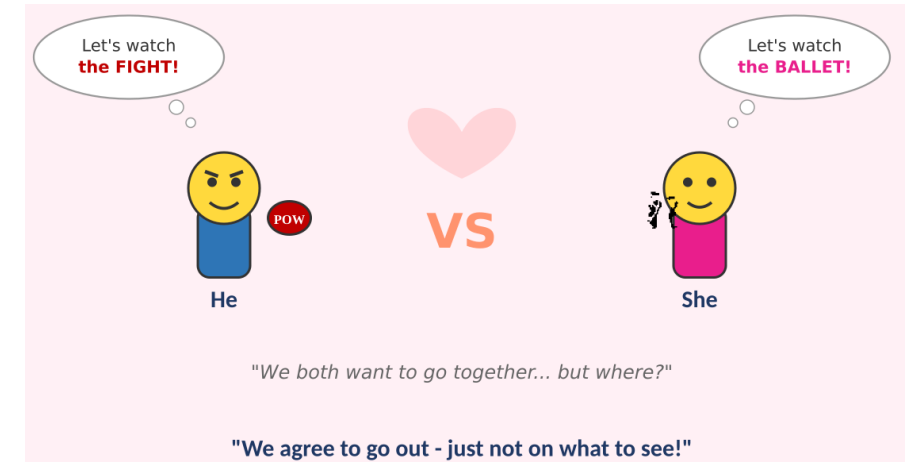
- 1) Both Left → best for both
- 2) Both Right → ok
- 3) Mismatch → nothing

Let's Play Games 3

Battle of Sexes

Red → Prize Fight Blue → Ballet

		Player B	
		Red	Blue
Player A	Red	(2, 1)	(0, 0)
	Blue	(0, 0)	(1, 2)



Think about the players' preferences

For Player A:

- 1) Both Prize Fight → A prefers
- 2) Both Ballet → B prefers
- 3) Mismatch → bad

Let's Play Games 4

Chicken / Hawk-Dove

Red → Aggressive Blue → Yield

		Player B	
		Red	Blue
Player A	Red	(0, 0)	(3, 1)
	Blue	(1, 3)	(2, 2)



Think about the players' preferences

For Player A:

- 1) A aggressive + B yield → best for A
- 2) Both yield → decent
- 3) A yield + B aggressive → worst for A
- 4) Both aggressive → disaster

What Just Happened?

You played **four different games.**

Each had a different **structure**, a different **tension**,
and probably led to a different classroom **outcome.**

Now let's understand why.

***02** How to Play Games?*

Finding Equilibrium

- Pre-Nash equilibrium
(Dominant strategy, iterative dominance, maximin strategy)
- Nash equilibrium
- Applications
(pricing games, rational pigs)

Equilibrium Concepts in Game Theory

There are multiple equilibrium concepts in game theory, each designed to capture a **different strategic property**. A useful game-theoretic solution concept should satisfy two key criteria:



Behavioral realism – it should reflect how rational players are **most likely to behave**.



Universality – it should exist across all games; otherwise, its predictive power is incomplete.



Pre-Nash Equilibrium

- *Dominant Strategy Equilibrium*
- *Iterated Dominance Equilibrium*
- *Maximin Strategy Equilibrium*



Nash's Concept

- *Nash Equilibrium*



Post-Nash Equilibrium

- Subgame-Perfect Nash Equilibrium
(in sequential game setting)

Dominant Strategy Equilibrium

Question: Would you confess or deny?

		Suspect B	
		Deny	Confess
Suspect A	Deny	1 yr, 1 yr = (2, 2)	10 yr, free = (0, 3)
	Confess	free, 10 yr = (3, 0)	5 yr, 5 yr = (1, 1)

Dominant Strategy

A strategy that gives a player better payoffs than all other strategies, *regardless of what the other player might choose*.

Confess dominates Deny for A / B
(confess, confess)

Dominant Strategy Equilibrium

Question: Would you confess or deny?

		Suspect B	
		Deny	Confess
Suspect A	Deny	1 yr, 1 yr = (2, 2)	10 yr, free = (0, 3)
	Confess	free, 10 yr = (3, 0)	5 yr, 5 yr = (1, 1)

Commandments on Dominance

- If you have a dominant strategy, use it.
- Expect your opponent to use hers if she has one.

They choose strategies that make both worse off, creating a dilemma — hence the name **Prisoner's Dilemma**.

- Each player has a dominant strategy
- The equilibrium is not jointly optimal

The central question: If cooperation is so obviously better, why don't players cooperate more often?

TV Quiz Show: “Shafted”

In the final round of SHAFTEd (a British quiz show), two finalists decide how to divide the prize money:

- If both choose “shaft,” neither wins anything.
- If one chooses “share” and the other “shaft,” the player who shafts takes all the money.
- If both choose “share,” they split the prize equally (£75,000 each).

		Player B	
		share	shaft
Player A	share	(50%, 50%)	(0%, 100%)
	shaft	(100%, 0%)	(0%, 0%)

“Their fate will be in each other's hands as they decide whether to share or to shaft”.



Iterated Dominance Equilibrium

- Recall: A dominant strategy equilibrium asks, “which strategy will definitely be played?”
- But some games have no dominant strategy equilibrium.
- Instead, we can ask: “Which strategies will definitely not be played?”
- A rational player will not choose a strategy that is always worse than another strategy, **regardless of** the opponent’s action.

Definition:

- By iteratively eliminating dominated strategies, we may — though not always — reach a unique outcome.
- This is called an **Iterated Dominance equilibrium** outcome.

Iterated Dominance Equilibrium

The diagram illustrates the process of iterated dominance equilibrium through three stages of strategy elimination:

Stage 1: Initial Payoff Matrix

		Firm B		
		Left	Center	Right
Firm A	Top	(3, 6)	(7, 4)	(10, 1)
	Middle	(5, 1)	(8, 2)	(14, 3)
	Bottom	(6, 0)	(6, 2)	(8, 5)

Note: The 'Top' row is crossed out with a purple line.

Stage 2: After Eliminating 'Top' for Firm A

		Firm B		
		Left	Center	Right
Firm A	Middle	(5, 1)	(8, 2)	(14, 3)
	Bottom	(6, 0)	(6, 2)	(8, 5)

Note: The 'Left' and 'Center' columns are crossed out with a purple line.

Stage 3: After Eliminating 'Left' and 'Center' for Firm B

		Firm B
		Right
Firm A	Middle	(14, 3)
	Bottom	(8, 5)

Note: The 'Bottom' row is crossed out with a purple line. The 'Middle' row and its payoff (14, 3) are highlighted with a red box.

- **(Middle, Right)** is the only outcome that survives the “**Iterated Dominance**”, thus **iterated dominance equilibrium**.
- *Note: iterated dominance does not always yield a unique outcome – which is why this equilibrium selection algorithm is not universal enough.*

Iterated Dominance Equilibrium

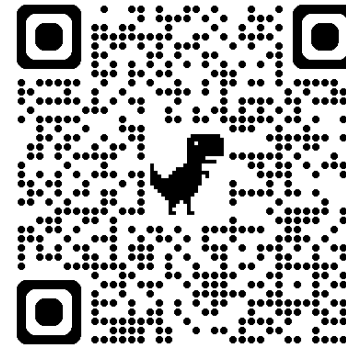
Game 1 P-test



Pick a Number in $[0, 100]$.

One who picked the number closest to **70% of the average** of all picks can win a gift.

Are you rational?



Game 2 Final Grade Bonus

**FREE
BONUS!**

Choose 5 free extra marks or 2 marks.

You get whichever extra marks you chose if **less than 20% of all students chose 5 marks**.

Otherwise, no one gets any extra marks.

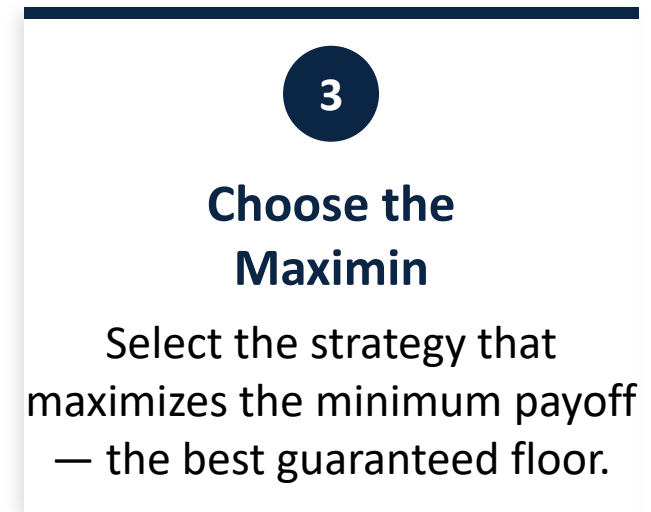
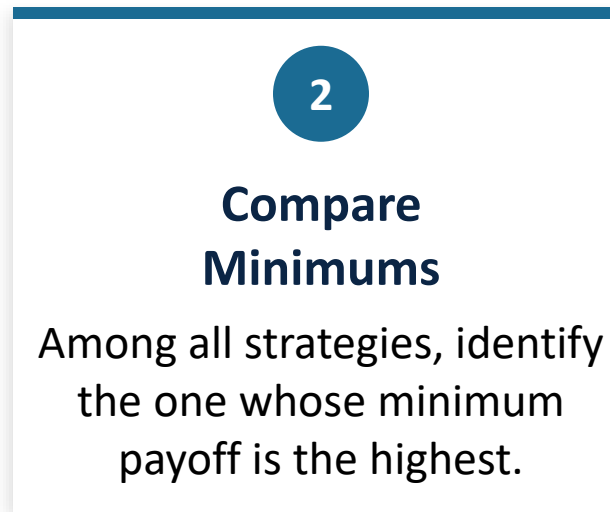
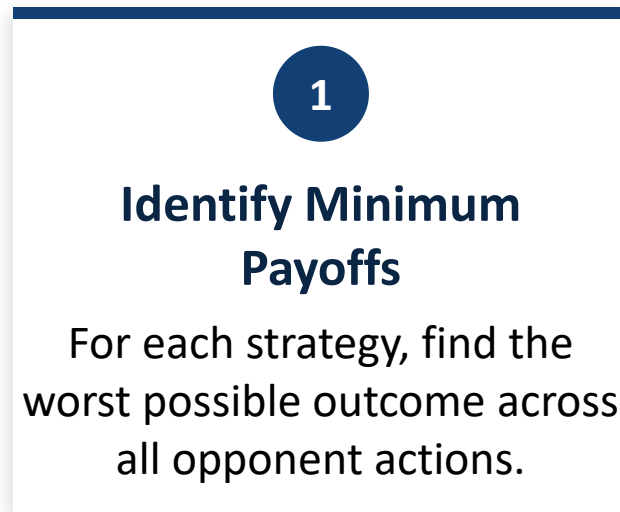
I Trick You or You Trick Me?

Strategic
Implication

Not all games have dominant or dominated strategies – the concept is **not universal enough**.

Maximin Strategy Equilibrium

What if the game contains a 'fatally' bad outcome and/or the opponent is **not always rational**?



Maximin = Maximize the Minimum payoff → Choose the strategy with the **best worst-case outcome**

Maximin Strategy Equilibrium

		Player B		A's Min
		Left	Right	
Player A	Top	(10, 4)	(8, 15)	8
	Bottom	(-100, 5)	(20, 10)	-100
B's Min		4	10	

For Player A, “**Top**” is his maximin strategy,

For player B, “**Right**” is his maximin strategy.

(Top, Right) is a maximin strategy equilibrium

If **both players are rational**, (Bottom Right) or (20, 10) would be the *iterated dominance equilibrium*.

Higher payoff, but requires trust

However, what if **player A doubts player B's rationality**? Outcome then would be (Top, Right) or (8, 15).

Lower payoff, but guaranteed floor

Maximin Strategy Equilibrium – Strategic Implications

Two Equilibria, Two Assumptions

If A trusts B's rationality

(Bottom, Right) $\rightarrow$ (20, 10)

Higher payoff, but requires trust

If A doubts B's rationality

(Top, Right) $\rightarrow$ (8, 15)

Lower payoff, but guaranteed floor



Key Implication: Given game structure, Player B might have an incentive to *fake her rationality*. Sometimes, you may want to create a fog for your own good!

Take-Away Points

You want your rivals to play their own maximin.

Brinkmanship — “Be Crazy like a Fox”



Maximin Strategy in Real Business

Nokia's Smartphone strategy was, after all, a maximin.

"Nokia is cornered by competitors and in need of a major transformation. Nokia is like a man standing on a burning oil platform who jumps into icy waters to escape the flames"

(Leaked memo by Stephen Elop, Nokia CEO in 2011)

"Nokia is a company that has internalized a culture of mediocrity"

(New York Times, 2012)

Micron's 2019 announcement to idle its production facility?

"We're just going through an air pocket. We will cut our monthly chip production by 5%"

(March 2019, Sanjay Mehrotra—CEO at Micron)



NOKIA

Micron

Skepticisms of the Maximin Concept

		Player B	
		Left	Right
Player A	Top	(10, 4)	(8, 15)
	Bottom	(-1, 5)	(20, 10)



1

Weak Justification for Rational Players

There is very little justification for a rational player to adopt maximin as an equilibrium concept. *How bad is “fatally bad”?* What if the downside for A was **-1** instead of **-100**?



2

Not Self-Enforcing

A maximin strategy equilibrium is not always a Nash equilibrium. *Players may have incentives to deviate once they observe the opponent's choice.*



3

Fragile as an Equilibrium Concept

The maximin concept may be acceptable for **risk-averse players** but represents **inconsistent and fragile behavior**. It is a conservative heuristic, not a robust solution concept.

Nash Equilibrium

Why we need Nash Equilibrium (NE)?

✗ Dominant Strategy

Not universal — many games have no dominant strategy for any player. Only works when one strategy always beats all others.

✗ Maximin Strategy

Fragile — works for risk-averse or conservative play, but not self-enforcing. Players may deviate once they observe the opponent's choice.

So what properties must a game equilibrium have?

John Nash's Answer



No player should have an incentive to deviate from the equilibrium, once it is reached.

*The equilibrium must be self-enforcing: each player's strategy is the **best response** to the other players' strategies.*

Nash Equilibrium

Definition

Nash Equilibrium is a **stable state** in which **no** player can gain anything by **deviating** from her chosen strategy, given that other players' strategies remain unchanged.

(Concept) That is, all players' strategy choices should be **self-enforcing**.

Suppose that there are two players A and B, each choosing strategy x and y , respectively. Then, for (x, y) to be a Nash equilibrium,

*x should be A's best response to B's choice of y , and
 y should be B's best response to A's choice of x .*

Thereby, no player has a reason to deviate from this Nash equilibrium.

Nash Equilibrium

Chicken Game

Two gangs race their cars toward each other. The first one to swerve loses face — but if neither swerves, both crash.



		Gang B	
		Straight	Swerve
Gang A	Straight	$(-100, -100)$	$(10, -2)$
	Swerve	$(-2, 10)$	$(0, 0)$

Key Observations

- No dominant strategy exists for either player.
- Dominance concept does not help us here. We need a different approach.

NE₁: (Straight, Swerve)

NE₂: (Swerve, Straight)

*Neither player
wants to deviate.*

Finding Nash Equilibria: Best Response

Best Response

A player's best response is the strategy that would give her the highest payoffs for all possible strategies chosen by other players.

1

Find A's Best Responses

For each of B's strategies, identify which strategy gives A the highest payoff. Mark these cells.



2

Find B's Best Responses

For each of A's strategies, identify which strategy gives B the highest payoff. Mark these cells.



3

Find the Overlap

Cells where BOTH players are playing their best response simultaneously are Nash equilibria.

Nash's Insight: Put yourself in your rival's shoes. Find the best response to each of their potential choices.

Exercise

Use dots (or underlines/circles) to mark each player's best response, then find where they overlap.

		Player 2				
		F	G	H	I	J
Player 1	A	<u>9</u> , <u>9</u>	7, 1	5, 6	3, 4	1, 1
	B	7, <u>8</u>	5, 2	3, 6	1, 4	3, 3
	C	5, 6	3, 3	1, <u>8</u>	<u>9</u> , 7	1, 5
	D	3, <u>9</u>	1, <u>9</u>	<u>9</u> , 4	7, <u>9</u>	<u>5</u> , <u>9</u>
	E	1, 2	<u>9</u> , <u>8</u>	7, 7	5, 6	3, 7

Instructions

Step 1: For each column, circle Player 1's highest payoff

Step 2: For each row, circle Player 2's highest payoff

Step 3: Cells with both circled = Nash Equilibrium

Three Nash Equilibria: (A, F) (E, G) (D, J) — all are self-enforcing.

if neither has an incentive to deviate

“A Beautiful Mind” and Nash Equilibrium



The Bar Scene

Five women walk in: four brunettes and a stunning blonde. John Nash gets the idea for Nash equilibrium.

Each of the four buddies starts forward to introduce himself to the blonde. Nash stops them, saying:

“If we all go for the blonde, we will all be rejected and none of the brunettes will talk to us afterwards because they will be offended. So let's go for the brunettes.”

Nash then runs to his professor to discuss his new idea.

Key Takeaway: Individual payoff maximization does not always lead to optimal results — but is the movie's “solution” self-enforcing?

The Challenge

Is the result in the movie actually a Nash equilibrium?

Why or why not? If not, what would be the Nash equilibrium for the situation?



Think about whether any single buddy **has an incentive to deviate** from Nash's proposed strategy. Would someone want to break ranks and go for the blonde?

\$ Application 1 – Prisoners' Dilemma in Pricing

Cooperation can increase the total payoff space. Why then don't players cooperate more often?

		Firm B	
		Price Low	Price High
Firm A	Price Low	(1, 1)	(3, 0)
	Price High	(0, 3)	(2, 2)

Dominant Strategy NE: **(Price Low, Price Low)**

Real-Life Price Wars

McDonald's–Burger King–Wendy's price war (2015): 10-piece nuggets at \$1.49

Amazon-Walmart: all-out price war in e-commerce since 2017

Price war among e-commerce is everywhere



The Dilemma: Both firms would be **better off** at (Price High, Price High), but each has an individual incentive to undercut — driving both to the **inferior** (Price Low, Price Low) equilibrium.

\$ Application 1 – Prisoners' Dilemma in Pricing

What should we do then?

A. Don't Get Caught in the First Place

- **Change the structure of the game** by making your product less sensitive to rival's prices.
- **Seek a competitive advantage:** Differentiate your product with design improvements, service, or advertising (e.g., Evian, iPhone).
- **Seek an unfilled niche:** *"Become a monopolist."*

B. Once Caught, Learn to Escape

- **Be creative: change the game structure** so that the mutually advantageous outcome is secured.
- **Direct communication (collusion)** is typically illegal
- **Communicate indirectly** via price leadership, price signaling, information agglomeration, strategic move

C. Repeated interaction may incentivize the players to play cooperatively.

\$ Application 1 – Prisoners' Dilemma in Pricing

Price War at 大学城南柳小吃街

价格表			
鹌鹑蛋一个	1元	鸭翅	5元
兰花干	3元	鸡肉肠	5元
豆干	3元	纯肉肠	5元
鸭心一串(3颗)	3元	面筋两串	5元
鱼豆腐	3元	鸭锁骨	6元
干张一条	3元	鸡腿	7元
土豆片一串	3元	鸭头	8元
干页豆腐	3元	鸭脖	10元
鸭爪	4元	鸭肠	10元
鸡爪	4元	鸭腿	10元
蔬菜两串	5元	兔头	18元
翅尖	5元	排骨	18元
鸭肝(3块)	5元	猪蹄	18元
鸭胗	5元	猪肘子	25元

Before April 2026

Market prices remained relatively stable



April 25 2026

A new entrant started a price war



May 10 2026

The incumbent stall responded by matching lower prices and joining the price war

\$ Application 1 – Prisoners' Dilemma in Pricing

Price War at 大学城南柳小吃街

A different story

B (new entrant) A (incumbent)

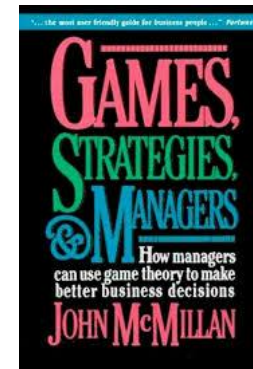
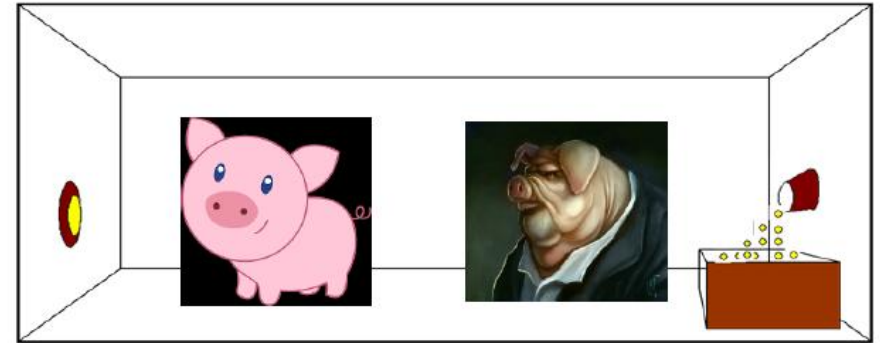


- Original Croissant Pricing
 - Stall A: 3.5 yuan
 - Stall B: 3 yuan
- Despite the lower price, Stall B attracted fewer customers. **Why?**
- A customer response: *"I always buy from Stall A and never from Stall B."*
- Suggests that consumer choice is influenced by factors beyond price, such as **habit**, **trust**, or **perceived quality**.

Application 2: Rational Pigs

The Setup

- Two “rational” pigs — one big (Napoleon) and one small (Pinkie) — are in a cage with a button at one end and a food dispenser at the other.
- Pressing the button dispenses **10g** of food, but the pig that pressed it must run to the dispenser while the other has already started eating.
- Napoleon, **being dominant**, **eats all remaining food once he arrives**.
- Pinkie's **only chance to eat** is by **reaching the food before Napoleon**, but must yield once he arrives. Pressing and running costs each pig **2g**.



Source: “Games,
Strategies & Managers”
(by John McMillan)



Application 2: Rational Pigs

Actions / Strategies	Payoff (Napoleon's, Pinkie's)
If Pinkie presses the button, Napoleon eats all 10g	(10, -2)
If Napoleon presses the button, Pinkie can eat 7g before Napoleon arrives	(1, 7)
If both press the button, Pinkie can run faster (given Napoleon's fat shape) and eat 3g	(5, 1)
If neither presses, no food of course	(0, 0)

Question: Who is going to press the button?



Application 2: Rational Pigs



	Press	Wait
Press	5, 1	<u>1</u> , <u>7</u>
Wait	<u>10</u> , -2	0, <u>0</u>

Who Presses the Button?

- **Pinkie's dominant strategy: Wait**
(Wait always gives Pinkie a higher payoff)
- **Napoleon's best response to Pinkie waiting: Press**
($10 > 0$, so Napoleon must do the work)
- **Nash Equilibrium: (Press, Wait) → (1, 7)**

Lesson: At times, a weakness can be a strategic strength and vice versa. — “Judo Economics”



Business Analogy: The OPEC Game

The Setup

- OPEC members agree to tight production quotas to keep oil prices high.
- Saudi Arabia (Napoleon) is the “stronger” player — large reserves, low production costs.
- Each smaller member has an incentive to cheat: overproduce beyond their quota to capture extra revenue at the supported oil price.

The Pig Game Analogy

Pig Game	OPEC
Napoleon (big)	Saudi Arabia
Pinkie (small)	Smaller OPEC members
Press button	Obey quota
Wait	Cheat (overproduce)

NE: SA obeys quota, others cheat — SA bears the cost of maintaining the cartel while small members free-ride.

If others cheat, SA still has incentive to maintain quotas — *its share is too large to risk a price collapse.*

SA's strength in market power may turn into weakness in strategic power. Being the “Napoleon” means *doing the work* while the “Pinkies” *free-ride.*

03

Sequential Game

Post-Nash equilibrium (Subgame Perfect Equilibrium)

Bargaining and negotiation

Credibility and threats

Sequential Games: What Changes?

We now extend our analysis to include “**dynamic**” – **sequential** decision making – features of the strategic decision.

Different timing

Players make strategic choices at different points in time

Multiple decision points

Players face several nodes where they must choose

Information structure

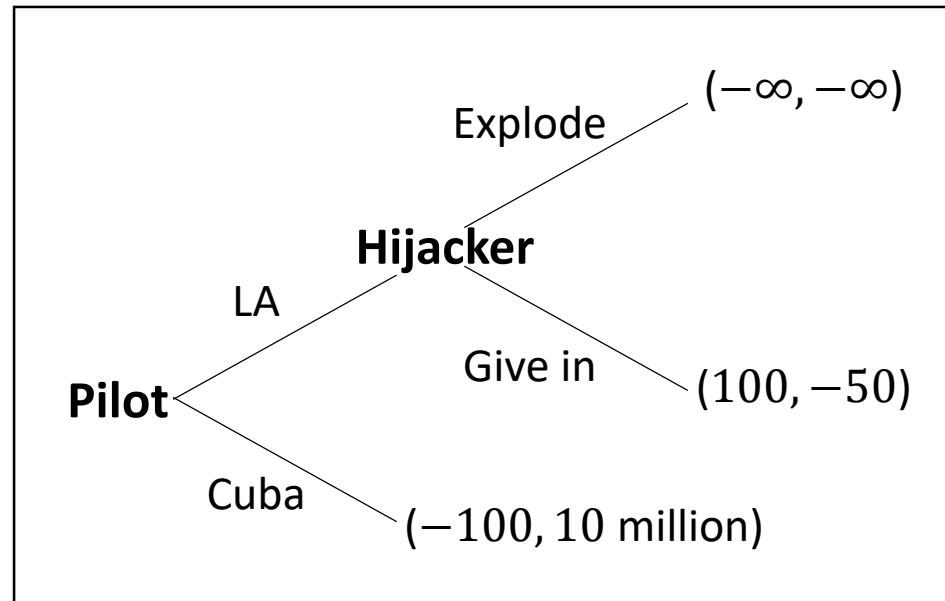
Players may observe earlier moves — distinguish actions from strategies

A strategy is a complete plan for every contingency — not just a single action.

The Hijacker's Dilemma

Scenario

- A hijacker wants the pilot to fly to Cuba
- The pilot must decide first:
 - *Comply and fly to Cuba*
 - *Refuse and land at LA*
- If the pilot lands at LA, the hijacker then decides whether to:
 - *Give in*
 - *Blow up the plane*



- The Hijacker threatens: “If you fly to LA, I will blow up the plane.”
- Is this threat credible? Should the pilot believe it? If you were the pilot, what would you do?

Nash Equilibrium in Sequential Games?

The hijacker's strategy is a **complete contingent plan** (what to do if pilot chooses LA, *and* what to do if pilot chooses Cuba).

	(Give In if LA, Give In if Cuba)	(Give In if LA, Explode if Cuba)	(Explode if LA, Give In if Cuba)	(Explode if LA, Explode if Cuba)
LA	100, -50	100, -50	$-\infty, -\infty$	$-\infty, -\infty$
Cuba	-100, 10m	$-\infty, -\infty$	-100, 10m	$-\infty, -\infty$

Issue: Which Nash equilibrium is more likely to occur?

- {Cuba, (Explode if LA, Give In if Cuba)} is a NE but a poor predictor.
- Reason: it relies on a **non-credible threat** by the hijacker.
- *Would the hijacker really blow up the plane?*

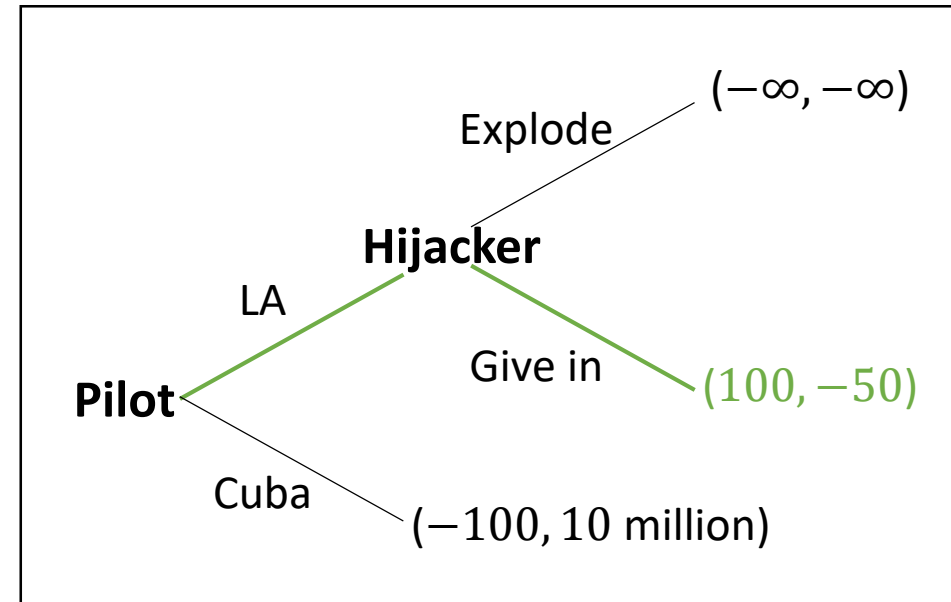
Backward Induction and SPNE

Any algorithm then to **sort out reasonable equilibrium** outcomes in sequential games?

Backward Induction

In finite dynamic games: We **start with the last period first** and work backwards through successive nodes until we reach the beginning of the game.

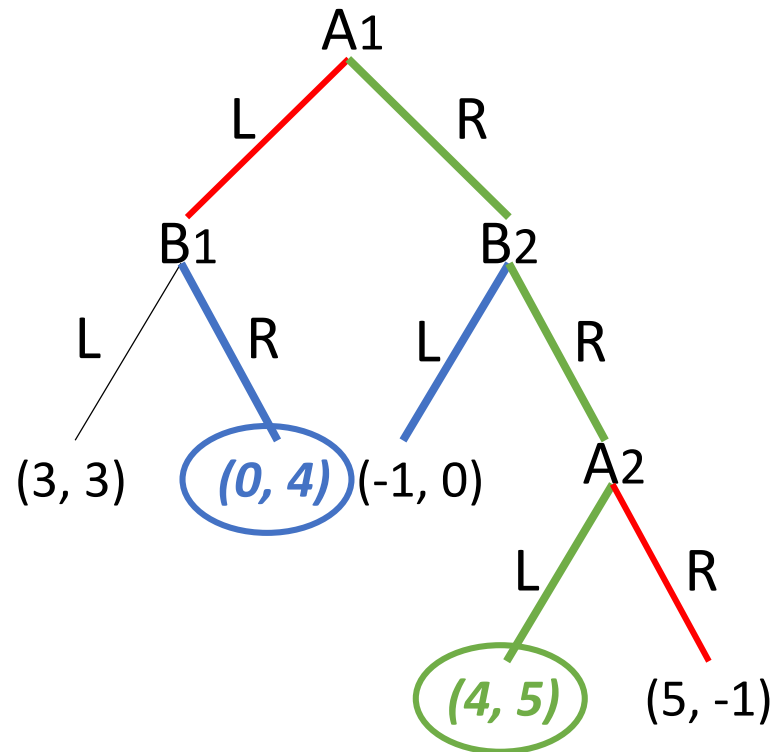
“Play forward but think backward!”



- At the last node: Hijacker prefers Give in (-50) over Explode ($-\infty$)
- **Knowing this**, Pilot chooses LA (100) over Cuba (-100)
- **(LA, Give in)** is the subgame perfect Nash equilibrium (SPNE)
- This way, we can **rule out empty threats or promise**.

Example of a **Non-credible** Promise

Payoffs are (A's, B's)

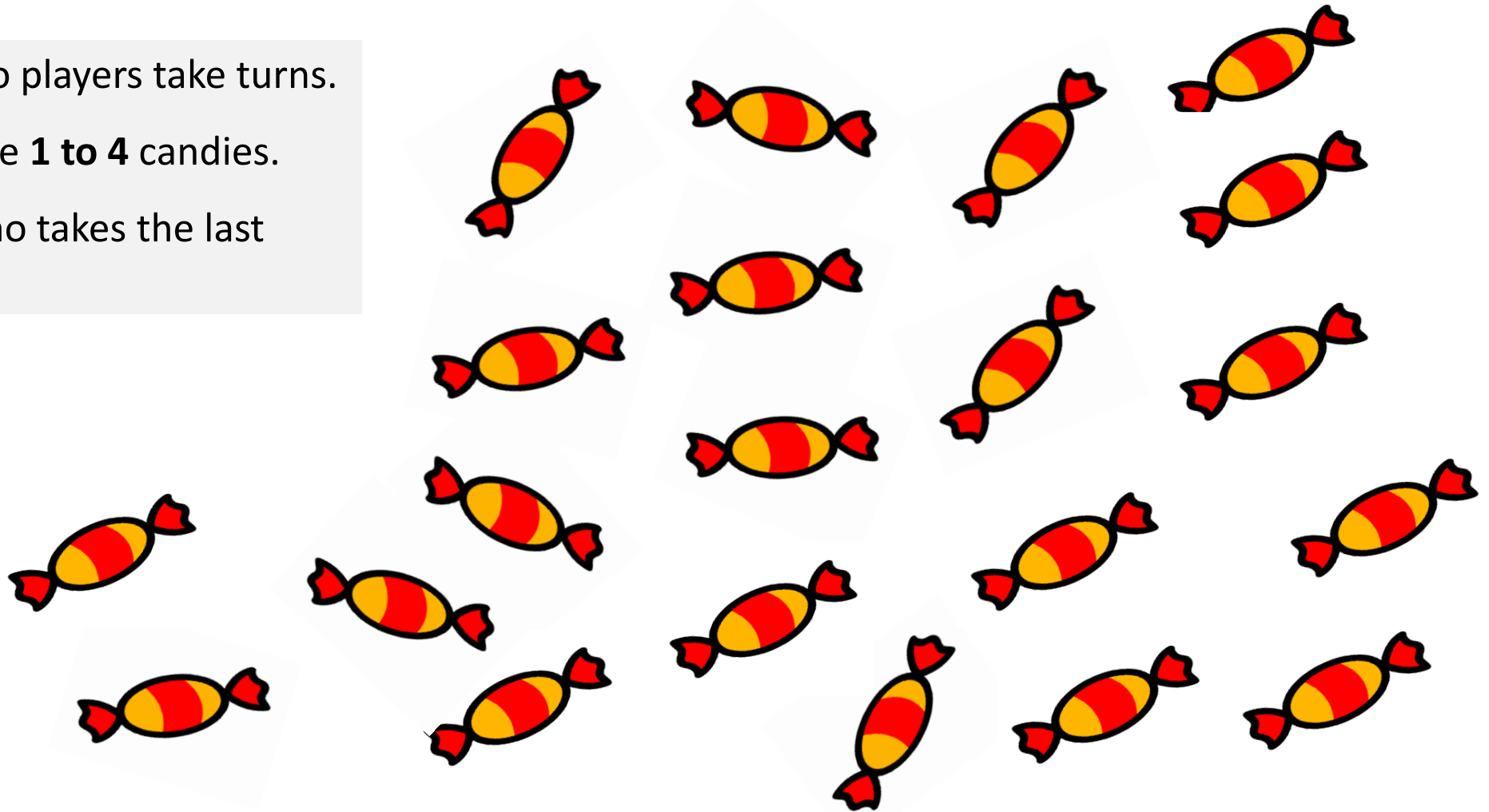


- (0,4) was the **unique** subgame perfect Nash equilibrium (SPNE). But, interestingly, both players can be better off at (4,5).
- What if A promises to B as below, as indicated by the **green arrows**?
"We both can be mutually happier by playing smarter. If you select R at B2, then I promise to select R at A1 and L at A2."
- It is **not credible**, thus not subgame perfect.

Experiment: First-Mover Advantage

21 Candies. Two players take turns.

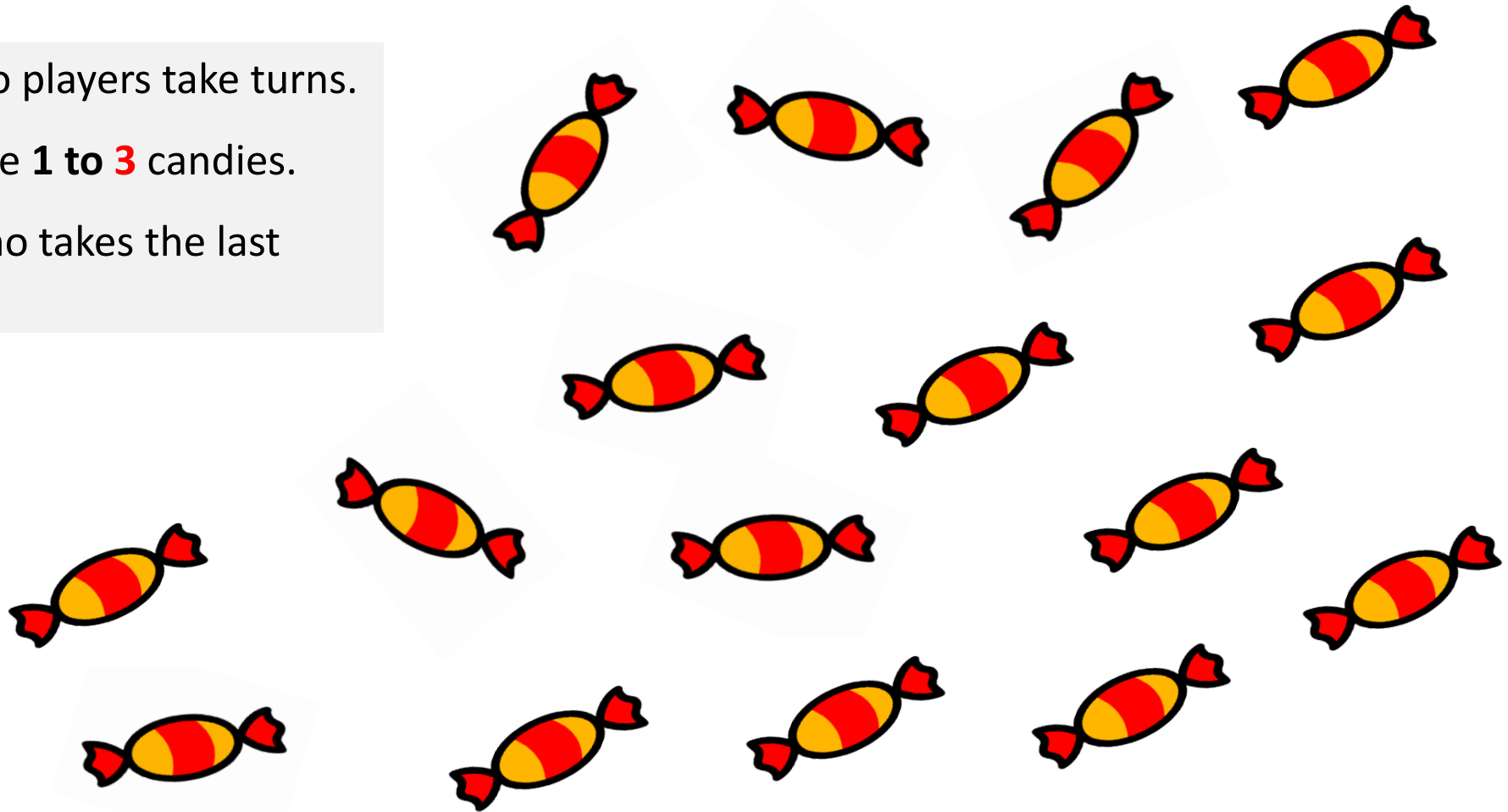
- Each turn, take **1 to 4** candies.
- The player who takes the last candy wins.



Experiment: First-Mover Advantage

16 Candies. Two players take turns.

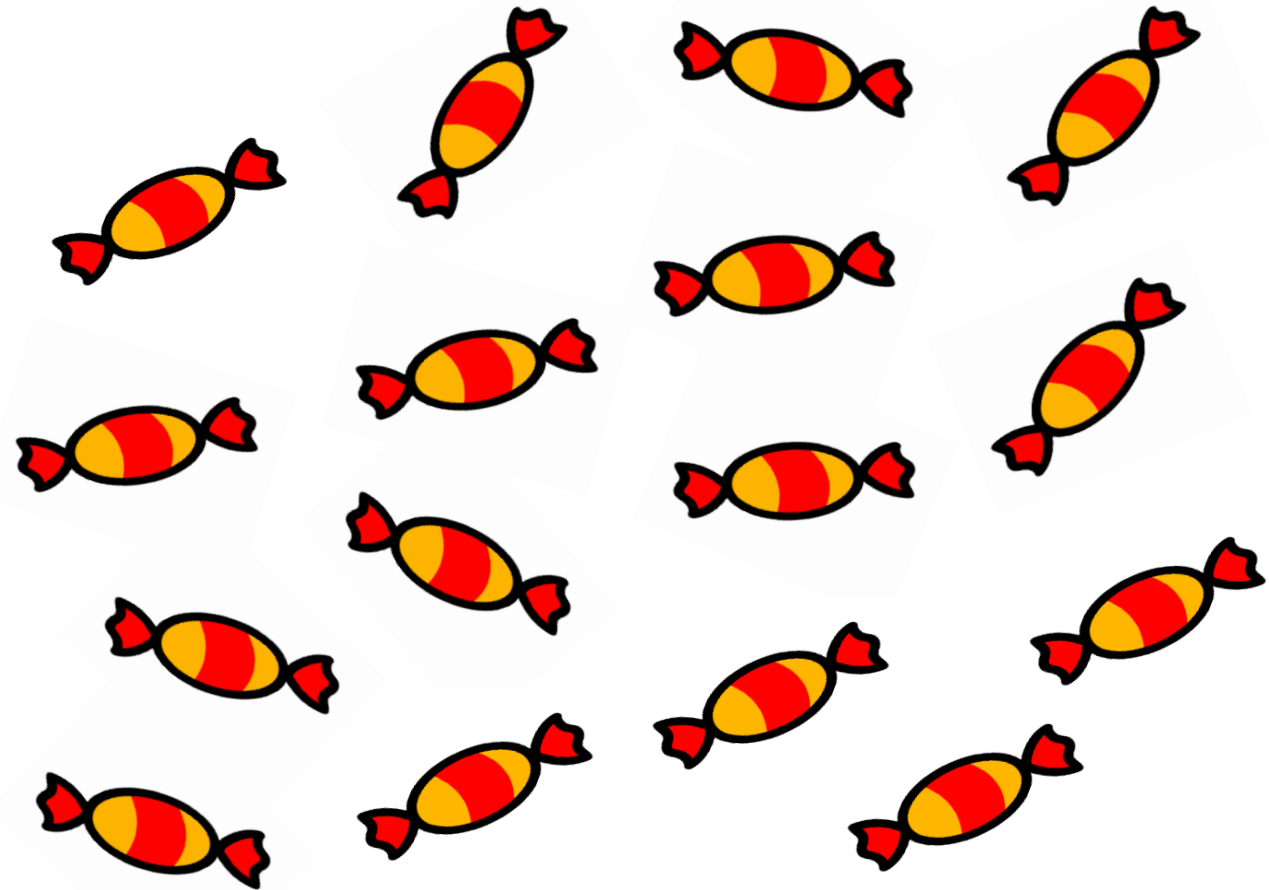
- Each turn, take **1 to 3** candies.
- The player who takes the last candy wins.



Experiment: Second-Mover Advantage

Two players

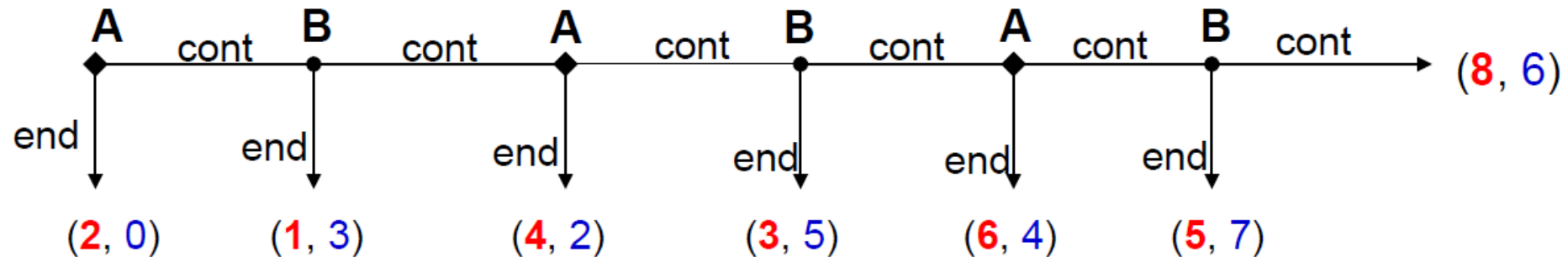
- Each turn, a player must take 1 to **3** candies
- The player who takes the last candy wins



Centipede Game

A and B alternate: each can Stop (end) or Continue. Payoffs grow but the stopper gets more.

(A's, B's)



How would you play the game if you were A? How about B?

Remarks on the Centipede Game

Theory

Backward induction
says: Stop immediately
(END in round 1).

Experiments

People actually chose
CONTINUE many times
rather than END.

Why?

Altruism, reputation-building,
bounded rationality,
misunderstanding, or expectations
about opponent's irrationality.

Lesson: In **repeated** business relationships, **short-run defection** may be theoretically rational, but **long-run cooperation** is often practically superior.

Tigers and Sheep

Story Setup

100 hungry tigers and 1 sheep on an island.

Rules:

- Tigers are perfectly rational
- Tigers prefer: eating > not eating > being eaten
- If a tiger eats, it becomes sleepy → vulnerable
- A sleepy tiger can be eaten by another tiger



Will any tiger eat the sheep?

Solution (Backward Induction)

- **Even number of tigers → no one eats**
- **Odd number → first tiger eats**
- *“Strategic restraint from pure self-interest.”*

Application: Bargaining

Story Setup

Two kids, **A** and **B**, are bargaining over **\$100** given by their grandpa.

Daddy sets the rules:

- Each kid can claim **only 0 or 50 or 100** dollars.
- If the sum of exceeds 100 → stalemate → penalty of \$1 from each kid's piggy bank for being greedy.

Why It Matters



Business Context:

Stalemates = opportunity cost from failed/delayed negotiations.

Common in: **Labor disputes, contract negotiations, merger talks**

Application: Bargaining

Simultaneous Bargaining

Kid A	Kid B		
	Claim 0	Claim 50	Claim 100
Claim 0	(0, 0)	(0, 50)	(0, 100)
Claim 50	(50, 0)	(50, 50)	(-1, -1)
Claim 100	(100, 0)	(-1, -1)	(-1, -1)

Three Nash Equilibria exist:

$(0, 100)$ → A claims all

$(50, 50)$ → Equal split

$(100, 0)$ → B claims all

💡 Key Insight: Coordination Problem

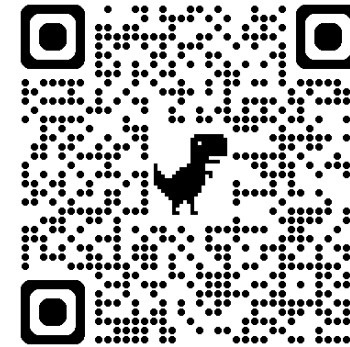
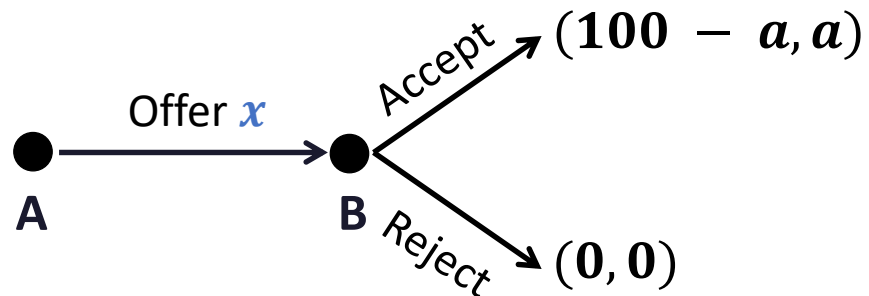
*Multiple equilibria mean players may fail to coordinate, leading to **inefficient outcomes**.*

Application: Bargaining

Ultimatum Sequential Bargaining – Take-it or Leave-it

Revised Story Setup

- **A** makes a take-it-or-leave-it offer of amount x to **B**.
- If **B** **accepts**, the deal is done.
- If **B** **rejects**, no one gets a dollar.
- Tie-break rule: “If indifferent between accept and reject, it is to be accepted.”



Subgame Perfect Nash Equilibrium (SPNE)

- **B's strategy**: Accept any offer $x \geq 0$ (something > nothing)
- **A's strategy**: Offer $x^* = 0$ (take everything)
- **Equilibrium payoff**: $(A's, B's) = (100, 0)$

In theory, the first mover takes all.

How about you?

Application: Bargaining

Ultimatum Sequential Bargaining – Take-it or Leave-it

What Experiments Show

Psychological Effects

***Altruism** gets in the way. **Mongolians** almost always offered 50:50, rejecting anything less than 50%.*

Reputation Effect

***B never accepted** A's offer of less than 30. B knew how to tame A.*

Inequity Aversion

*People **prefer fairness** and resist inequality. May explain why absolute terms matter less than relative shares.*

Does Scale Matter?

*Would it change if the game was over **\$10,000 instead of \$100**?*

Application: Bargaining

Alternating Offer **Sequential Bargaining – Take-it or counteroffer**

Revised Story Setup

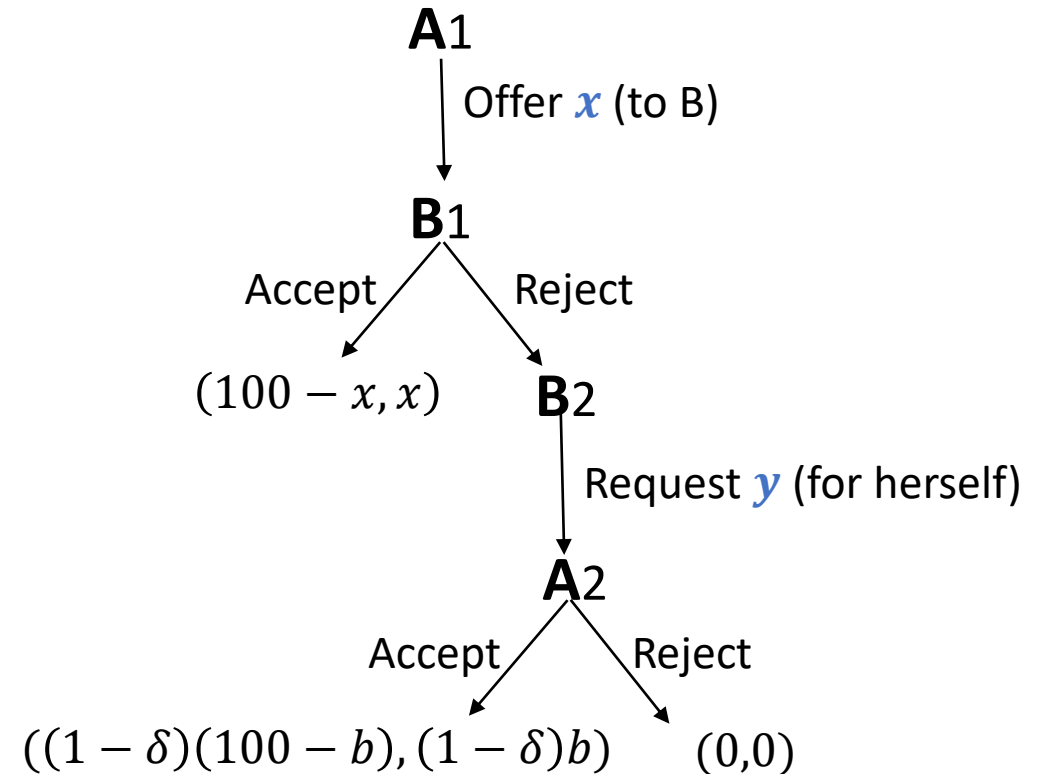
Stage 1: **A** offer x to **B**. **B** accepts or rejects.

Stage 2:

- If **B accepts**, game is over $\rightarrow (100 - x, x)$
- If **B rejects**, **B** asks for y (counteroffers $100 - y$ to **A**)
 - If **A accepts** $\rightarrow \{(1 - \delta)(100 - b), (1 - \delta)b\}$
 - If **A rejects** $\rightarrow (0,0)$

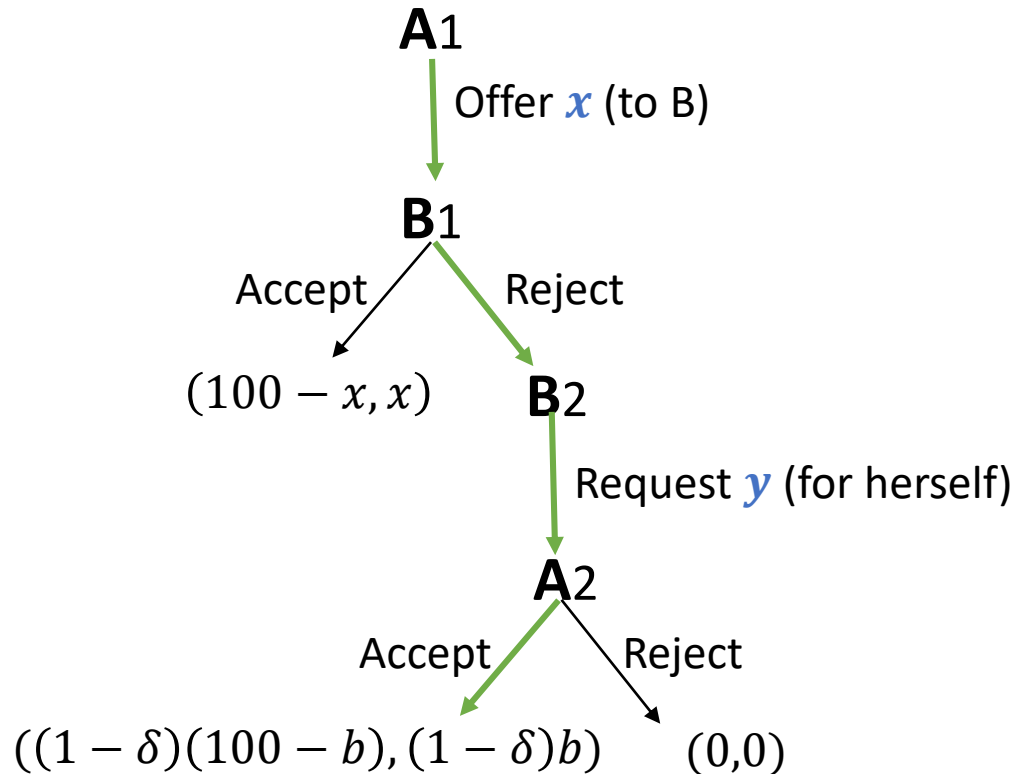
$0 < \delta < 1$ means time has value.

(Candies melt down! Dollars depreciate!)



Application: Bargaining

Alternating Offer *Sequential Bargaining – Take-it or counteroffer*



Subgame Perfect Nash Equilibrium (SPNE)

- At A2: A accepts any request $y \leq 100$ since $(1 - \delta)(100 - y) \geq 0$. Otherwise, A rejects.
- At B2: B asks for $y^* = 100 \rightarrow$ B can secure **$100(1 - \delta)$** for herself
- At B1: B accepts any offer $x \geq 100(1 - \delta)$, rejects otherwise. (Note, if B rejects, A's payoff = 0.)
- At A1: A offers $x^* = 100(1 - \delta)$ to B.

Equilibrium payoff

$$(A's, B's) = (100\delta, 100(1 - \delta))$$

Application: Bargaining

Implications: Credible Threat and Its Limits

Take-it-or-leave-it

- B gets nothing. $(A's, B's) = (100, 0)$
- B could only say yes or no — no effective threat.

Take-it-or-counteroffers

- B can reject and counteroffer.
- $(A's, B's) = (100\delta, 100(1 - \delta))$

B's rejection at B1 is a **credible threat** to A.

- B can secure $100(1 - \delta)$ by rejecting A's initial offer.
- But the threat is limited — B must forgo 100δ to exercise it.
- The smaller δ , the more bargaining power B has.

Check: What if $\delta = 1$? What if $\delta = 0$?

Application: Bargaining

Who Has the Advantage?

First Mover Wins

- If $\delta > 50\%$: First-mover advantage
- On a hot day, candies melt fast $\rightarrow$ B's bargaining edge gets smaller.

Second Mover Wins

- If $\delta < 50\%$: Second-mover advantage
- On a cold day, B can afford to wait $\rightarrow$ B's edge gets bigger.

Strategic Point

- A wants to **increase** δ (raise delay cost).
- B wants to **decrease** δ (lower delay cost).

Application: Bargaining

Reality Check: Why Do Delays⁶⁶ Happen?

Theory Says: Settle Early

- Rational players reach early agreement.
- When delayed, both lose.
- Labor negotiations: late agreements come at the cost of strikes
- Civil disputes: “settle if you can, otherwise only lawyers get rich”

Reality Says: It Happens

Why do agreements get delayed in reality?

- Asymmetric expectations or ungrounded optimism
- Reputation-building incentive
- Emotional factors override rational calculation

Lesson: Delays are not recommendable. The value of agreement shrinks with time.

Application: Bargaining

Daddy's Wisdom: Mechanism Design

“Hey, kids. You had better reach an agreement in the first round. Otherwise, I will take a half!” (i.e., set $\delta = 50\%$)

Resulting Equilibrium: (A's, B's) = (50, 50)
Perfectly fair division!

Why it works:

1. Setting $\delta = 50\%$ balances bargaining power between first and second mover
2. Neither side can gain from delaying — half is lost immediately
3. This is mechanism design: designing rules to achieve desired outcomes

04

Strategic Moves

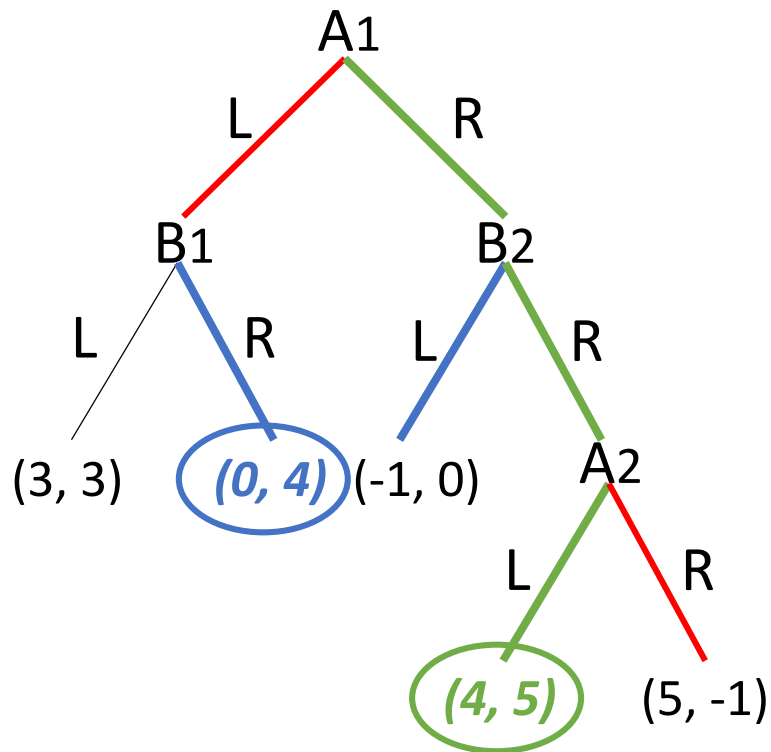
Commitment

Credibility

Information Asymmetry

Motivation: Why Credibility Matters

Payoffs are (A's, B's)



SPNE outcome: (0, 4)

But both could reach (4, 5) — a better outcome for both players.

A's promise: "If you select R at B2, I promise to select R at A1 and L at A2."

→ **Not credible**. A would deviate at A2.

Challenge: If you were player A, what would you do to realize (4, 5)?

- Cheap talk is not enough
- **Credibility is the key!**

Credibility, Commitment, and Strategic Move

You want to influence competitors' choices in your favor — via threats or promises.
But talk is cheap! You need to make it credible.

A strategic move = an action that influences others' expectations about your behavior, securing your advantage.

How commitment works:

It limits or modifies your own options

Rivals know you are committed

If credible, rivals change their strategy to your benefit

Thomas Schelling

2005 Nobel Prize

First introduced the strategic aspect of commitment
— the idea of 'Strategic Moves.'

Brinkmanship as a Strategic Move/Commitment

Brinkmanship: pushing a situation to the verge of disaster to force the rival to concede.

This is achieved by creating the impression that you are willing to use extreme methods rather than concede.

Classic Examples

Historical:

- Cold War: Mutual Assured Destruction (MAD)
- Cortés burning ships (1519)
- Castle moats: keeping your own soldiers from retreating

Business Examples

Modern Business:

- Trump's tariff escalation (245%)
- Capacity pre-commitment
- 'Nuclear option' in negotiations

The Madman

Key insight:

Unpredictability and perceived irrationality can be strategic advantages.

Nixon's 'Madman Theory'

Chicken Game: Strategic Moves

		Gang B	
		Straight	Swerve
Gang A	Straight	$(-100, -100)$	$(10, -2)$
	Swerve	$(-2, 10)$	$(0, 0)$



- How many (pure strategy) Nash equilibria?

(Swerve, Straight)

(Straight, Swerve)

- What actually happens in reality?

It is hard to say. Under Nash theory, either of the two equilibrium points is an equally 'rational' outcome.

Challenge: How can you assure your win, then?

Recall: Implications of Maximin Strategy

		Player B		A's Min
		Left	Right	
Player A	Top	(10, 4)	(8, 15)	8
	Bottom	(-100, 5)	(20, 10)	-100
B's Min		4	10	

For Player A, “**Top**” is his maximin strategy,

For player B, “**Right**” is his maximin strategy.

- 1 **(Top, Right)** is a maximin strategy equilibrium
- 2 **(Bottom, Right)** is an iterative dominance equilibrium

Note that B's payoff is higher under 2 than 1.

- You play your maximin strategy if you want to secure a certain bottom line payoff.
- Ironically, by **forcing your rivals to play their own maximin strategy**, you can have a better outcome. Thus, you remain unpredictable!

Winning the Chicken Game

You are "Gang A"

(i) Create your own fog

- Put a "weirdo" in the driver seat
- An irrational — or rationally irrational — or unpredictable player may have the upper hand
- **Recall:** You want to force rivals to play their maximin strategy



What if there is a 50:50 chance for Gang A to come straight or swerve?

		Gang B	
		Straight	Swerve
Gang A	Straight	(-100, -100)	(10, -2)
	Swerve	(-2, 10)	(0, 0)

(ii) Add sticks / Reduce temptation

- Show no mercy to a chicken (Mafia?)
– "Life as a chicken is like no life..."
- Make the tempting option (swerving) less attractive by reducing your own payoff from them: **dis-incentivize**.



What if Gang A is concerned about his boss' brutality, which is known to Gang B as well?

		Gang B	
		Straight	Swerve
Gang A	Straight	(-100, -100)	(10, -2)
	Swerve	(-101, 10)	(0, 0)

Winning the Chicken Game

You are "Gang A"

(iii) Incentivize: Add carrots

- "Die for the honor of my family." (Or, put a suicidal guy in the drive seat)
- The key is making the winning option more attractive (by increasing your own payoff to it – incentive)



(iv) Remove your own temptation

- Tie up or pull out your steering wheel.
- The key is getting rid of the options you might be tempted to play.



What if Gang A was suicidal and Gang B knows about it?

		Gang B	
		Straight	Swerve
Gang A	Straight	(-1, -100)	(10, -2)
	Swerve	(-2, 10)	(0, 0)

What if "Swerve" is not even an option for Gang A?

		Gang B	
		Straight	Swerve
Gang A	Straight	(-100, -100)	(10, -2)
	Swerve	(-101, 10)	(0, 0)

Strategic Moves in the Pricing PD

How to escape when explicit collusion is illegal?

		Firm B	
		Price Low	Price High
Firm A	Price Low	(40, 40)	(65, 35)
	Price High	(35, 65)	(60, 60)

Dominant Strategy Equilibrium → (40, 40)

Cooperation → (60, 60)

Three strategic moves:

- I. Cross-Shareholding
- II. Most Favored Customer (MFC) Clause —
Promise to rivals via customers
- III. Price Matching Guarantee (PMG) —
Threat to rivals via customers

I. Cross-Shareholding

What if each firm acquires 20% of the other?

		Firm B	
		Price Low	Price High
Firm A	Price Low	(40, 40)	(65, 35)
	Price High	(35, 65)	(60, 60)



		Firm B	
		Price Low	Price High
Firm A	Price Low	(40, 40)	(59, 41)
	Price High	(41, 59)	(60, 60)

$$\text{Firm 1: } \Pi_1 = 65 \times 0.8 + 35 \times 0.2 = 59$$

$$\text{Firm 2: } \Pi_2 = 35 \times 0.8 + 65 \times 0.2 = 41$$

- New equilibrium: (High, High). PD solved!
- **Strategic Lesson:** Internalize the rival's loss into yours and your gain into the rival's.

II. Most Favored Customer (MFC) Clause

Use customers as hostage through a Promise

“I will not lower the price. **If I ever offer a lower price** to any other customer, I will offer it to you – the MFC customers – as well”

Logic: MFC makes your own low-pricing costly → you commit to keeping prices high → rival follows suit.

Rationale:

- Removes own incentive to cut price
- Implicit promise to rivals
- Customers become the hostage

In Real Business:

- Popular in the online business rivalry
- Mixed results on competition
- Antitrust authorities are alert

III. Price Matching Guarantee (PMG)

Use customers as hostage through a Threat to rivals

“If **other firms lower the price**, we will match it right away.”

Logic: PMG makes rival's price cuts pointless → rival commits to high prices → cooperation emerges.

Rationale:

- Eliminates rival's incentive to cut
- Implicit threat: I'll match instantly
- Punishes any price-cutter

In Real Business:

- All major British supermarkets (since 2014)
- 'Intensity of price war is abating' (FT, 2015)
- Amazon's Product Ads = info agglomeration

The Irony of Strategic Moves

MFC and PMG exploit the same conditions that make price competition brutal:

- Customer price awareness
- Customer price sensitivity
- Low switching costs

And these same conditions can lead to cooperation — if a strategic move works!

Promise vs Threat

MFC = Promise

Fat Cat Effect: commit to being soft → rival also becomes soft

PMG = Threat

Credible threat: any price cut is instantly matched → no gain from cutting

Information Asymmetry: The Market for Lemons

Akerlof (1970, Nobel 2001)

Sellers know quality, buyers don't

→ Adverse selection

→ High-quality sellers exit

→ Only 'lemons' remain

→ Market unravels

"Bad money drives out good" (Gresham's Law)

Solutions:

- Warranties
- Certifications
- Reputation systems

IS Connection: Platforms reduce asymmetry

- Guazi used cars
- Alibaba trust scores
- Carfax vehicle history

Signaling: Overcoming Information Asymmetry

Spence (1973, Nobel 2001)

The informed party takes a costly action to credibly reveal private information.

Platform connections

- Verified reviews
- Seller badges
- Quality certification

"Cheap talk won't work — just like cheap threats."

"If you have quality, you need a credible way to show it."

Education

Signal of ability, not just learning

Warranties

Signal of product quality

IPO Underpricing

Signal of firm value

Game Theory Meets IS Strategy

IS Concept	Game Structure	Key Strategic Lever
Platform competition	PD / Coordination	Standards, network effects
Market entry (digital)	Sequential game	Credibility of threats
Network effects	Coordination (Pareto eq.)	First-mover, tipping point
Data as strategic asset	Information asymmetry	Signaling, adverse selection
Algorithmic pricing	Repeated PD	Tacit collusion risk
Platform ecosystems	Multi-game overlay	PD + BoS + Chicken

Game theory adds a competitive and interactive lens to the IS frameworks from Days 1–4.

The Strategic Thinking Checklist

Rationality

Are you rational? Is your opponent?

Preferences

What do players want? How do they rank outcomes?

Sequence

Simultaneous or sequential? Who moves first?

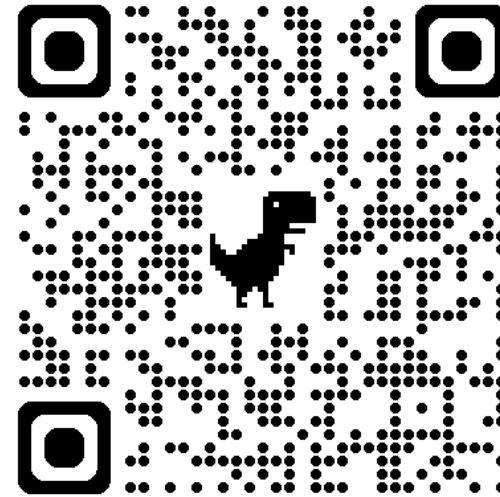
Conflict of Interest

Aligned or opposing goals? Trust? Coordination?

Information Structure

What do you know? What does your opponent know?

Q&A



dchen@kean.edu CBPM B214

Office hours: Tue & Thu 11:15–12:30 | Wed 14:00–16:30